

LensCNN: closing the sim-to-real gap for CNN Einstein-radius estimation with a realistic simulation generator

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ABSTRACT

Deep learning promises fast parameter estimation for the $\sim 10^5$ strong gravitational lenses expected from *Euclid* and the Roman Space Telescope, yet convolutional neural networks (CNNs) trained on simplified simulations typically degrade on real data — a “sim-to-real gap”. We show that *physical realism in the training simulations*, rather than network architecture or domain adaptation, is the decisive factor in closing this gap for Einstein-radius (θ_E) regression. Our generator assembles real ingredients in physical units: real HST galaxies provide the lens light, with each deflector’s mass modelled as a singular isothermal sphere whose scale is fixed by that galaxy’s measured SDSS velocity dispersion ($\sigma_v \rightarrow \theta_E$); real COSMOS sources are ray-traced through this mass; and real empty-sky backdrops and an empirical, focus-diverse PSF supply correlated noise and instrument response. The defining feature is observational self-consistency: one real galaxy supplies both a system’s light and its mass. We train a scale-conditioned CNN with a Gaussian negative-log-likelihood head. On a frozen spectroscopic-lensing benchmark of 62 SLACS and 40 S4TM lenses (Bolton et al. 2008 *b_{SIE}*), the native-HST model reaches $R^2 = 0.64$ (SLACS) and $R^2 = 0.90$ (S4TM), and $R^2 = 0.71$ on the *Euclid*-degraded benchmark, matching conventional modelling at a fraction of the amortized inference cost. Systematic ablations show real background structure is the single most important ingredient, while linking each deflector’s light and mass is the decisive step. The predicted uncertainty correlates with true error (Spearman $\rho = 0.6$ – 0.7) and gates a confident subsample to a 3–6% failure rate.

Key words: gravitational lensing: strong – methods: data analysis – techniques: image processing – galaxies: elliptical and lenticular, cD

1 INTRODUCTION

1.1 Strong lensing and the Einstein radius

Strong gravitational lensing occurs when a massive foreground galaxy (the deflector) lies close to the line of sight to a background galaxy (the source), bending and magnifying the source light into arcs or Einstein rings. Formally the geometry is set by the lens equation

$$\vec{\beta} = \vec{\theta} - \vec{\alpha}(\vec{\theta}), \quad (1)$$

which maps a source position $\vec{\beta}$ to its image positions $\vec{\theta}$ through the scaled deflection angle $\vec{\alpha}$ produced by the deflector’s projected mass. Modelling a lens therefore requires three ingredients: the *lens light* (the deflector’s surface brightness, well described by a Sérsic profile), the *lens mass* (the projected mass that generates $\vec{\alpha}$), and the *source light* (the background galaxy). For a singular isothermal sphere (SIS) the characteristic scale is the Einstein radius

$$\theta_E = 4\pi \frac{\sigma_v^2}{c^2} \frac{D_{ls}}{D_s}, \quad (2)$$

where σ_v is the deflector velocity dispersion and D_{ls}, D_s are galaxy-diameter distances. We adopt the SIS/SIE description

throughout and detail how the mass is constructed for training in Section 3.4. θ_E is a fundamental observable: it constrains galaxy mass-density profiles, the dark-matter content of early-type galaxies, and H_0 via time-delay cosmography (Meneghetti 2021, Sah; Treu 2010). Wide-field surveys — *Euclid* (Laureijs et al. 2011), Roman (Spergel et al. 2015), and LSST (Ivezić et al. 2019) — are expected to find $\sim 10^5$ lenses.

1.2 The need for automated modelling and the sim-to-real gap

Traditional modelling fits parametric profiles by likelihood-based inference (MCMC, nested sampling); it is accurate but costly (~ 3 min per lens; Ivezić et al. Cao) and needs expert supervision. CNNs amortize this: once trained they estimate θ_E from an image in milliseconds (Hezaveh et al. 2017; Perreault Levasseur et al. 2017). The obstacle is the *sim-to-real gap*: CNNs trained on simplified simulations degrade on real data, manifesting as prior-pull (collapse to the training mean), slope compression, and overconfident, miscalibrated uncertainties. The physical origins of this gap are not yet well characterised. Domain-adaptation studies (Perreault Levasseur, Hezaveh & Wechsler Cîp; Cîp Aga; Aga Swi) have so far been simulation-to-simulation, deferring real-data validation; Swi (Bus) evaluate on real *Euclid* lenses but train on fully parametric simulations and report degraded real-vs-simulated performance.

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1.3 This work

We present a CNN θ_E estimator trained on a physically self-consistent, real-ingredient pipeline. The intuition is simple: a neural network is only as good as the examples it learns from, so if those examples are cartoon simulations the network fails on real telescope images. We therefore build each training example around a *real* galaxy — using its true light, and giving it a mass fixed by that same galaxy’s own measured internal motion (its velocity dispersion) — so the network practises on systems that obey the same physics as the real lenses it must ultimately measure. Concretely, our contributions are:

- (i) **Observational self-consistency (GEN4):** each training system uses one real HST galaxy for both its light and its mass (θ_E from that galaxy’s measured σ_v ; Section 3.4), embedding the luminosity–mass link human modellers rely on when arcs are faint.
- (ii) **Real ingredients throughout:** real COSMOS sources, real empty-sky HST backdrops (correlated noise, field neighbours), and an empirical focus-diverse PSF, all in physical $\text{e}^- \text{s}^{-1}$.
- (iii) **Causal realism ablations:** a controlled decomposition of which ingredient closes the gap.
- (iv) **Domain-aware uncertainty:** a Gaussian-NLL head whose σ correlates with real-world error and acts as a confidence gate.
- (v) **Comprehensive real-lens validation** on a frozen spectroscopic benchmark (62 SLACS, 40 S4TM; Bolton b_{SIE}), its *Euclid*-degraded version, real *Euclid* Q1 lenses, and Roman simulations.

Throughout we distinguish two generations of the generator: **GEN4** reproduces the low-redshift HST domain (the SLACS/S4TM benchmark), while **GEN5** extends the pipeline to the higher-redshift *Euclid* and Roman domains. Our scope here is deliberately narrow — a single band and the single most robust parameter, θ_E — which we argue is the right first target for closing the gap (Section 6). This paper is organised as follows: Section 2 reviews related work, Section 3 the generator, Section 4 the network, Section 5 the results, Section 6 limitations and future work, and Section 7 concludes.

2 RELATED WORK

2.1 CNNs for strong-lens parameter estimation

Hezaveh et al. (2017) first applied CNNs to SIE-parameter recovery from simulated HST-like images; Perreault Levasseur et al. (2017) added Bayesian uncertainties. Both were simulation-only. Bus (Gaw) trained on HSC ground-based simulations with real empty-cutout injection — a recipe close to our hybrid approach — reaching 10–20% θ_E accuracy on 182 real HSC lenses, but with ground truth from their own YattaLens pipeline. Because an automated pipeline’s output is itself a model estimate that can share systematic biases with the network under test, agreement with it measures consistency rather than physical accuracy; we instead adopt the Bolton b_{SIE} radii, derived from SLACS spectroscopy and HST imaging independently of any lens-finding CNN, as a physically anchored benchmark. Gaw (Sch) compared a ResNet to conventional modelling on 31 HSC lenses, finding good θ_E agreement for $\theta_E \gtrsim 2''$ but poor shear recovery. Swi (Bus) (LEMON) is closest to our work: trained on 80k fully parametric *Euclid* VIS simulations, it is evaluated on Euclidised-HST and real *Euclid* Q1 lenses. Ivezić et al. (Cao) (TinyLensGPU) is a GPU-accelerated conventional pipeline with nested sampling and full posteriors; it provides per-lens θ_E and uncertainties and is our conventional-modelling reference.

2.2 Domain adaptation

Unsupervised domain adaptation for lens regression (Perreault Levasseur, Hezaveh & Wechsler Cip; Cip Aga; Aga Swi) improves target-domain accuracy and posterior coverage, but all published studies are simulation-to-simulation and defer real-data validation. Sim-to-real adaptation on real spectroscopic ground truth remains unclaimed; we find (Section 5) that domain adaptation does not improve over realism-calibrated simulations.

2.3 Positioning

Table 1 positions this work. Our differentiators are observational self-consistency, causal realism ablations, an independent spectroscopic ground truth, comprehensive cross-domain real-lens validation, and a domain-aware uncertainty that also functions as a practical confidence gate. We stress a reproducibility caveat on LEMON (Section 5.7): we cannot recover LEMON’s published aggregate metrics by re-scoring its released predictions against the literature θ_E , so we report our own transparent re-scoring separately from LEMON’s reported numbers.

3 DATA AND SIMULATOR

3.1 The frozen real benchmark

We evaluate on a frozen benchmark of 62 SLACS and 40 S4TM grade-A lenses observed with HST ACS/WFC in F814W, with ground-truth θ_E taken as the SIE Einstein radius b_{SIE} of Bolton et al. (2008) — a spectroscopic-lensing measurement independent of our pipeline. One SLACS system (J0955+0101) is excluded for a corrupted cutout. Cutouts are 128×128 px at $0.05''/\text{px}$ from MAST drizzled products. The benchmark is never used for training or model selection; it is touched only at logged evaluation checkpoints. As a harder test we also use 322 real *Euclid* Q1 lenses with PyAutoLens SIE ground truth. Table 2 summarises the sample. *No benchmark galaxy’s light stamp appears in the training lens-light library* (Section 3.3): a sky-position cross-match of the library against the benchmark finds zero galaxies within $5''$ of any benchmark lens (nearest separation $2798''$) and no name overlap, removing a potential leak-age path.

3.2 Source galaxies (real COSMOS)

Sources are real galaxies from the COSMOS_23.5 catalogue (56 062 GREAT3-vetted galaxies; Auger, Treu, Bolton, Gavazzi, Koopmans, Marshall, Moustakas & Burles Man), ray-traced through the lens mass by PALTAS (Man Wag) on LENSTRONOMY. An effective-surface-brightness cut, $\text{SB}_{\text{eff}} = m_{\text{auto}} + 2.5 \log_{10}(2\pi(r_{\text{flux}})^2) \leq \text{SB}_{\text{max}}$, retains $\text{SB}_{\text{max}} = 21.0$ for the native-HST arm and 22.5 for the *Euclid* arm; compact-source cuts remove unresolved objects. Source flux is renormalised to the measured SLACS source-magnitude distribution (mean ≈ 24.3 at $z \approx 0.65$; Wag New). For the high-redshift GEN5 arm the renormalised source is additionally dimmed to its drawn z_s with a luminosity-distance plus flat- f_y K -correction.

3.3 Lens-light library (real HST galaxies)

The lens light comes from real HST F814W stamps of non-lens early-type galaxies drawn from the SLACS/S4TM parent samples and, for GEN5, expanded by a footprint cross-match of HST frames against

Table 1. Comparison of representative strong-lens θ_E methods. “Real GT” indicates validation against ground truth independent of the method under test; “UQ” indicates per-lens uncertainty quantification. **[TODO: confirm ticks with final numbers.]**

Feature	LEMON	Gawade	HOLISMOKES X	Cao (TinyLens)	This work
Approach	CNN	CNN	CNN/ResNet	conventional	CNN
Training simulations	parametric	hybrid	—	—	real-ingredient
Real-lens GT	mixed	pipeline	pipeline	spectroscopic	spectroscopic (Bolton)
Realism ablations	no	no	no	n/a	yes
Uncertainty (UQ)	yes	no	no	yes	yes
Inference speed	fast	fast	fast	~3 min	fast

Table 2. Frozen benchmark sample properties. θ_E ranges are from Bolton et al. (2008) (b_{SIE} ; the value we adopt as ground truth); redshift and deflector-magnitude ranges are from the SLACS/S4TM survey papers.

Property	SLACS (62)	S4TM (40)
θ_E [arcsec]	0.69–1.78 (med. 1.17)	0.54–1.62 (med. 1.00)
z_l	~0.05–0.5	≤0.4
z_s	~0.2–1.2	~0.2–1.1
Deflector mag (F814W)	[TODO: compute]	[TODO: compute]

the SDSS σ_v pool. The core GEN4 results use 139 stamps; the GEN5 extension uses 488 (`g1b_kinematics_v1.csv`). Each stamp carries a measured σ_v , lens redshift z_l , apparent magnitude, effective radius, axis ratio q , position angle, and isophote-shape coefficients. Each stamp is injected at its native amplitude, with no brightness rescaling. We refer to this deliberately as a *lens-light* library: the light is real, but the mass is modelled (Section 3.4).

3.4 Lens-mass model and physical self-consistency (GEN4)

A galaxy’s mass — unlike its light — is not directly observed, so every training deflector is given a *model* mass built from its own measured kinematics rather than an independently drawn Einstein radius. We adopt an SIS: θ_E is fixed by the galaxy’s SDSS σ_v through Equation 2, evaluated at the drawn (z_l, z_s) ; no independent θ_E prior exists anywhere in the pipeline. Mass ellipticity and orientation are inherited from the measured light shape with empirical scatter ($q_{\text{mass}} = q_{\text{light}} \oplus 0.08$; $\Delta\text{PA} \sim N(0, 10^\circ)$), rotating mass and light together. For GEN5, isophote-anchored $m = 3, 4$ multipoles are added as perturbations on this isothermal base (amplitude set so the convergence-contour deviation equals the measured light-isophote deviation), so the profile remains fundamentally isothermal.

Because θ_E is derived from the deflector’s own luminosity-linked σ_v , the Faber–Jackson channel is present by construction: the network can read a faint deflector as a small σ_v and hence a small θ_E , exactly as a human modeller does when the arc is weak — the ingredient absent from parametric simulators, where θ_E is drawn independently of the light. A tempered importance-sampling scheme (weight $\propto (1/\text{density})^{0.6}$) keeps the effective θ_E distribution wide over $[0.45, 2.30]''$ while preserving a real light–mass correlation. We disclose the tension that perfect θ_E flatness and a fully preserved Faber–Jackson slope are incompatible for a σ_v library of finite width; the tempered prior is the compromise (simulator $\rho(\text{mag}, \theta_E) = -0.15$ vs real SLACS -0.32 : sign preserved, amplitude diluted).

Reliability of the $\sigma_v \rightarrow \theta_E$ label. SLACS analyses find early-type mass profiles close to isothermal on average (Koopmans et al. 2006; Auger et al. 2010), justifying the SIS mapping, but the σ_v scaling is only a loose predictor of the true Einstein radius: on the benchmark, $\theta_E^{\text{SIS}}(\sigma_v, z_l, z_s)$ reproduces Bolton b_{SIE} with a scatter of $0.20''$

(NMAD; 17% fractional) and a $-0.11''$ bias (Section 5.2, Fig. 2; using σ_v, z_l, z_s from Bolton et al. 2008). Crucially, this is *not* a floor on the network: the training arcs are rendered exactly at each galaxy’s θ_E^{SIS} , so there is no per-image label noise, and σ_v merely sets the training θ_E prior and a luminosity-based fallback. On real lenses the network reads the arc geometry directly and is *not* as tight as the σ_v relation (Section 5.2), so self-consistency on the real data is most where the arcs are faint (Bolton, Burles, Koopmans, Treu, Gavazzi, Moustakas, Wayth & Schlegel 2008). **[TODO: confirm ticks with final numbers.]**

3.5 Instrument realism, hybrid assembly, and survey operators

All components are generated in one physical system ($e^- s^{-1}$, F814W AB zeropoint 25.94) on a common $0.05''$ grid. The PSF is a real, focus-diverse empirical HST ePSF (not a Gaussian/Moffat); a held-out, benchmark-disjoint ePSF pool is used in an ablation to rule out PSF-mediated leakage. The native image is a physical sum in $e^- s^{-1}$: PALTAS source render + real deflector stamp + real empty-sky HST cutout (correlated noise, field neighbours) + real field companions + a Gaussian sky top-up matched to the real SLACS sky-RMS. A deterministic instrument operator degrades this to *Euclid* VIS (flux to I_E , ACS $\rightarrow$ VIS PSF-matching with the real Q1 kernel, 2×2 bin to 100 mas/px, Poisson noise at EWS depth with the real Q1 sky variance) and to the Roman WFI format. Every dataset version passes automated gates (sky-RMS ratio, lens peak/sky band, θ_E range/flatness, radial-profile overlay, and a Faber–Jackson gate) before scale-up.

4 CNN ARCHITECTURE AND TRAINING

The network is a scale-conditioned residual CNN: four residual stages, global average pooling, and a scale scalar (the pixel scale in arcsec/px) concatenated before an MLP head; an InceptionNeXt-stem variant is used for the Roman ensemble. The single band is arsinh-compressed ($a = 1.0$) and standardised per image, removing the absolute flux level (appropriate because θ_E is geometric) while the pixel scale enters as a conditioning scalar. The identical normalisation is applied to simulations and to the real benchmark. The head predicts a mean and log-variance ($\mu, \log \sigma^2$) trained with the Gaussian negative-log-likelihood, giving a per-image aleatoric uncertainty. Training uses Adam with a cosine schedule for 40 epochs at batch size 64, with the full dihedral group of flips and 90° rotations as augmentation. Model selection is on the simulation validation split only — deflector-, PSF-, and seed-disjoint from training — with the benchmark reserved for logged checkpoints. We train an ensemble of 16 networks (ResNet, InceptionNeXt; random seeds) and pre-register a small (2–3 member) ensemble on validation; headline numbers use the pre-registered ensemble (cnv2_3 for SLACS, r50_3 for S4TM). At evaluation we apply $8 \times$ dihedral test-time augmentation and report a total variance combining the mean aleatoric variance over views with the epistemic variance from view spread. Un-

Table 3. Training hyperparameters.

Setting	Value
Backbone(s)	ResNet / InceptionNeXt-stem
Input	1 band, 128×128 , $\text{arcsinh}+z\text{-score}$
Conditioning	pixel scale (arcsec/px)
Head	Gaussian NLL (μ , $\log \sigma^2$)
Optimiser	Adam, cosine LR, 40 epochs, batch 64
Augmentation	dihedral ($8 \times$ flips/rotations)
Ensemble	16 members; 2–3 pre-registered
Test-time aug.	$8 \times$ dihedral

Table 4. Two-domain performance of the GEN4 self-consistent model with 68% bootstrap CIs on R^2 and failure rate. A prediction is a failure if $|\text{frac. err}| > 15\%$. SLACS = full ensemble; S4TM = pre-registered member r50_3.

Domain	Sample	Bias''	RMSE''	NMAD''	R^2	Fail
Native HST	SLACS 62	+0.005	0.152	0.048	+0.64	15%
Native HST	S4TM 40	+0.013	0.088	0.047	+0.90	8%
Euclid-like	SLACS 62	−0.010	0.137	0.056	+0.71	15%
Euclid-like	S4TM 40	+0.027	0.117	0.082	+0.81	22%

certainty recalibration is described in Section 5.5. Hyperparameters are listed in Table 3.

Training and inference cost. Once trained, the network estimates θ_E from an image in a single forward pass (of order milliseconds per lens on a GPU). The training cost is amortised across the survey; the speed advantage over conventional forward modelling (≈ 3 min per lens; Ivezic et al. Cao) is realised at inference, the regime relevant to the $\sim 10^5$ lenses expected from *Euclid*. We note that conventional modelling also returns uncertainties and additional parameters (Section 5.6), so speed is not the only axis of comparison.

5 RESULTS

5.1 Headline benchmark performance

Table 4 reports the GEN4 self-consistent model on both domains. Figure 1 shows predicted vs true θ_E : the m3 baseline scatters far from the 1:1 line, while the real-unit self-consistent recipe tracks it. All metrics carry 68% confidence intervals from 10^4 bootstrap resamples of the lens sample (Appendix ??); on the small benchmarks these are wide (SLACS $R^2 = 0.64$, CI $[0.45, 0.82]$; S4TM $R^2 = 0.90$, CI $[0.86, 0.93]$), which we report explicitly rather than quote point estimates alone. We note for transparency that the SLACS headline is the full pre-registered ensemble, whereas the S4TM headline is its best-performing pre-registered member (r50_3, $R^2 = 0.90$; the full ensemble gives $R^2 = 0.81$); the member was fixed on validation before the benchmark was touched.

The project baseline (“m3”), a CNN trained on simplified forward-operator simulations, fails on the benchmark: $R^2 = -1.02$ (SLACS, 55% catastrophic) and -0.53 (S4TM). Its signed error sweeps from positive at small true θ_E to negative at large, crossing zero near the training-mean attractor ($\sim 0.95\text{--}1.0''$) — the signature of prior-pull. Per-feature Spearman correlations are weak ($|\rho| < 0.3$), so the failure is systemic to the training distribution, not architectural. On the real *Euclid* Q1 benchmark the GEN4 model reaches bias -0.104 , NMAD 0.094 , $R^2 = +0.609$, fail 32%; the GEN5 high-redshift model and post-domain-adaptation results are [TODO: pending (E5, E6)].

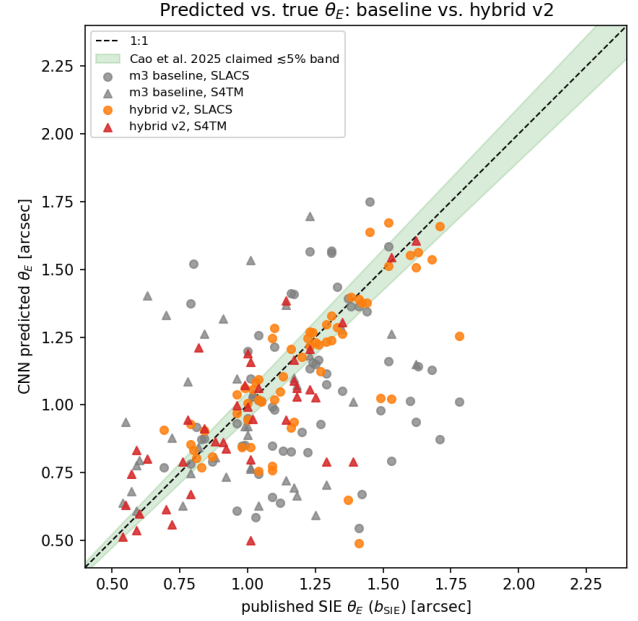


Figure 1. Predicted vs true Einstein radius θ_E for the SLACS and S4TM benchmark lenses. The m3 baseline (grey) collapses toward the training mean (SLACS $R^2 = -1.02$), while the real-unit self-consistent recipe (orange/red) tracks the 1:1 line within the Ivezic et al. (Cao) $\leq 5\%$ band. [TODO: regenerate with the final GEN4 ensemble; current panel shows the pre-self-consistency hybrid recipe.]

5.2 The network reads lensing geometry, not a σ_v proxy

A natural concern is that our results merely reflect the luminosity– σ_v correlation baked into the training labels. We test this directly. Taking the measured SDSS velocity dispersion and redshifts of the benchmark lenses (from Bolton et al. 2008), the SIS prediction $\theta_E^{\text{SIS}} = 4\pi(\sigma_v/c)^2 D_{ls}/D_s$ reproduces the true b_{SIE} with NMAD $0.20''$ (17%), $R^2 = +0.05$ (Pearson $r = 0.68$), and a $-0.11''$ bias (Figure 2, open symbols; $N = 58$). The CNN on the identical lenses is *four times tighter* (NMAD $0.05''$, $R^2 = 0.64$; filled symbols). Had the network only learned the σ_v – θ_E proxy it would be capped at $\sim 17\%$; instead it reads the arc configuration directly. The self-consistency mechanism thus sets the training prior and supplies a luminosity fallback for faint arcs, but the network’s precision on well-resolved arcs is geometric.

5.3 Domain of validity

The training θ_E support is $[0.45, 2.30]''$; real lenses above the ceiling are systematically under-predicted (e.g. 2/5 COSMOS lenses in Section 5.7). We report the out-of-support fraction per sample [TODO: (compute for SLACS-62, S4TM-40, Euclid Q1)] and restrict quantitative claims to the in-support regime.

5.4 Causal ablations

Table 5 and Figure 3 remove exactly one ingredient at a time from the real-unit hybrid recipe (native SLACS $R^2 = +0.27$), under an identical protocol. Physical self-consistency (Section 5.6) is then a separate, larger step on top ($+0.27 \rightarrow +0.64$); [TODO: re-run the two dominant ablations on the GEN4 recipe to confirm the ladder

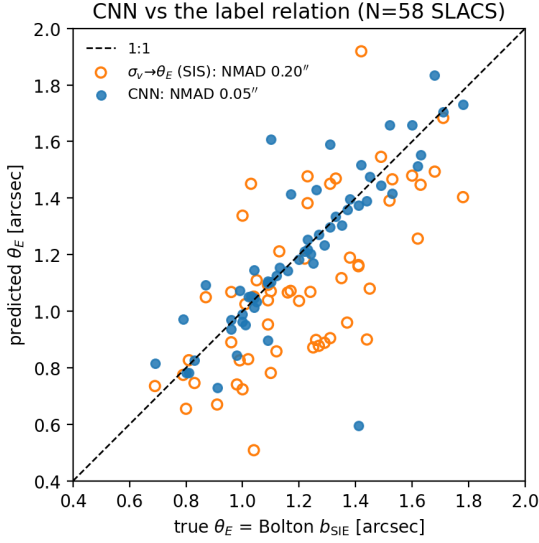


Figure 2. The CNN (filled) versus the $\sigma_v \rightarrow \theta_E$ SIS label relation (open) against the true b_{SIE} on the SLACS benchmark. The network tracks the 1:1 line four times more tightly than the σ_v relation (NMAD 0.05'' vs 0.20''), showing it reads the lensing geometry rather than a luminosity- σ_v proxy. Generated by `analysis/make_labelnoise_figure.py`; σ_v, z_l, z_s from Bolton et al. (2008).

Table 5. Ablations: each variant removes one ingredient from the hybrid recipe. R^2 /fail and median fractional error per sample.

Variant	SLACS R^2 /fail	S4TM R^2 /fail	SLACS med.	S4TM med.
Full recipe (v2)	+0.27/23%	+0.47/38%	-2.6%	-1.8%
Skewed θ_E prior	-0.52/29%	-0.03/50%	-6.3%	-7.8%
Gaussian PSF	-0.46/31%	+0.06/50%	-5.4%	-9.5%
Gaussian noise	-5.10/58%	-14.67/90%	+24.1%	+99.5%
Broad ePSF pool	+0.22/21%	+0.42/38%	-1.6%	-2.2%

(E8).] Real empty-sky backdrops dominate by an order of magnitude: trained on uncorrelated Gaussian noise, the network reads real correlated background and field neighbours as lensed flux and over-predicts catastrophically (S4TM median +99.5%). The flat θ_E prior and the empirical PSF are each individually necessary (removing either flips full-sample SLACS R^2 negative), and the benchmark-matched ePSF pool adds nothing over a fully disjoint archival pool, clearing the leakage concern.

5.5 Uncertainty and confidence gating

Table 7 reports the calibration and gating of the Gaussian-NLL head. The Spearman correlation between predicted uncertainty and absolute fractional error is positive for both samples, so the model flags its own failures. Table 6 and Figure 4 give the full failure-rate vs retained-fraction trade-off obtained by gating on σ : retaining the most confident 50% drops the SLACS failure rate from 16% to 3%, and the most confident 25% contains no failures for either sample. We caution that the *raw* uncertainty is over-confident on real data (Figure 5): the empirical 1σ coverage is 40% (SLACS) and 60% (S4TM) against a nominal 68%, so we apply a single global σ -recalibration (scale factor $k \approx 2.0$ for SLACS, 1.4 for S4TM) before quoting calibrated intervals. A 50%-throwaway gate is not itself a survey strategy;

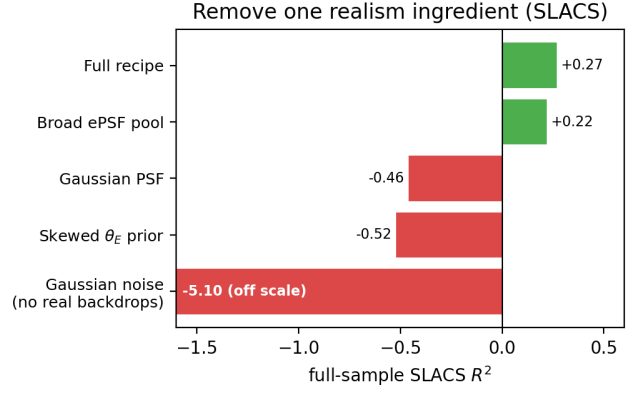


Figure 3. Ablation of the real-unit hybrid recipe (full-sample SLACS R^2). Removing the real empty-sky backdrops (“Gaussian noise”) is catastrophic and off-scale ($R^2 = -5.10$), identifying real background structure as the dominant ingredient; removing the flat prior or the empirical PSF each flips R^2 negative; the benchmark-disjoint (“broad”) ePSF pool nearly matches the full recipe, ruling out PSF-mediated leakage.

Table 6. Confidence-gating trade-off: failure rate among the most confident retained fraction (gating on predicted σ). From `analysis/calibration_gating.py`.

Retain most confident	SLACS fail	S4TM fail
25%	0.0%	0.0%
50%	3.1%	0.0%
75%	8.5%	3.3%
100%	15.9%	7.5%

Table 7. Uncertainty calibration and confidence-gated performance (GEN4).

Sample	Spearman ρ	med. σ/μ	full fail	conf.-half fail
SLACS (62)	+0.57	3.0%	15%	3%
S4TM (40)	+0.70	3.0%	8%	0%

the point is that the σ ranking is informative, and the trade-off curve lets a survey choose an operating point.

5.6 Comparison to conventional modelling (Cao)

Using the per-lens predictions kindly provided by Ivezić et al. (Cao), we compare our CNN to TinyLensGPU on the identical SLACS lenses, both against Bolton b_{SIE} . [TODO: Table of head-to-head bias/NMAD/ R^2 /fail (§5.4).] Our CNN matches the accuracy class of conventional modelling at a much lower amortised inference cost. We emphasise a fair comparison: TinyLensGPU also delivers per-lens uncertainties and full posteriors and is not prohibitively expensive on modern hardware, so our advantage is amortized throughput, not a categorical capability gap. [TODO: Add a CPU/GPU wall-clock comparison (E7).] Physical self-consistency lifts native SLACS R^2 from +0.27 (hybrid recipe) to +0.64 (GEN4), the single largest step in the ladder.

5.7 Comparison to LEMON

We evaluate both methods on the exact lenses LEMON reports predictions for, in the Euclidianised-HST domain, scoring each subsample

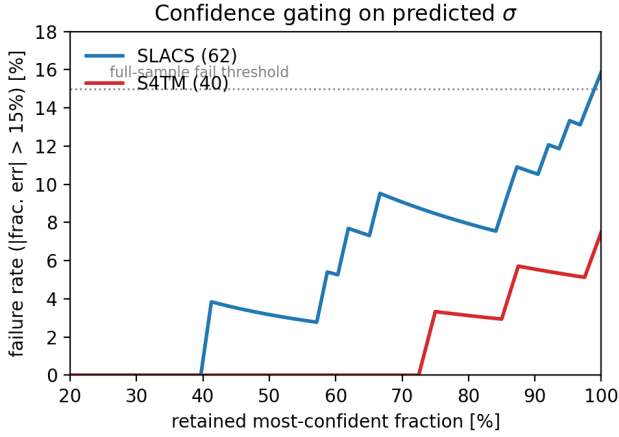


Figure 4. Confidence-gating trade-off: failure rate ($|\text{frac. err}| > 15\%$) among the most-confident retained fraction, gating on the predicted σ . The most confident $\sim 40\%$ (SLACS) and $\sim 70\%$ (S4TM) contain no failures; the curves reach the full-sample rates (dotted line: 15%) only when the least-confident lenses are included. Generated by `analysis/make_gating_figure.py`.

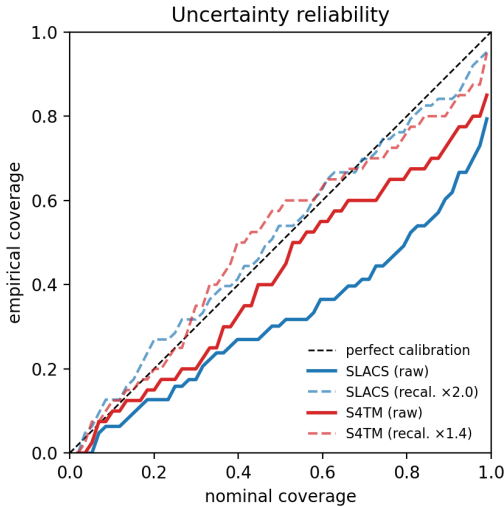


Figure 5. Uncertainty reliability: empirical vs nominal coverage. The raw predicted σ is over-confident (solid curves below the diagonal); a single global rescaling ($\times 2.0$ for SLACS, $\times 1.4$ for S4TM), fit so the 1σ coverage matches, restores calibration (dashed curves on the diagonal). Generated by `analysis/make_calibration_figure.py`.

against its own independent published ground truth (Table 8). On every subsample that has a real published θ_E we lead decisively: on SLACS-29 (Bolton b_{SIE}) we reach $R^2 = +0.57$ where LEMON collapses to $R^2 = -4.26$ (LEMON over-predicts by $\sim +0.29''$ mean bias), and on the EEL sample we again lead by a wide margin. COSMOS-5 is a small- N wash where both struggle — we under-predict 2/5 lenses that lie above our θ_E ceiling ($2.3''$, out-of-support and disclosed). The ACS-13 subsample has *no* published θ_E — only an arc radius (Koopmans, Treu, Bolton, Burles & Moustakas Paw) — so we exclude it from the θ_E aggregate (unlike LEMON’s headline, which folds 13/60 arc-radius systems into a single “ θ_E ” number); scored against the arc radius for transparency, our scatter is nonetheless tighter (NMAD $0.31''$ vs $0.36''$), while we carry the expected

Table 8. Head-to-head with LEMON on its own released predictions, per subsample, each scored against its independent published ground truth (Euclidised-HST domain). “cat.” is the catastrophic-outlier fraction. ACS-13 has no θ_E ground truth (arc radius only) and is excluded from the θ_E aggregate.

Subsample (GT)	Metric	Ours	LEMON
SLACS-29 (Bolton b_{SIE})	R^2	+0.57	−4.26
	NMAD''	0.045	0.307
	cat.	7%	55%
EELs-12 (Oldham 2017)	R^2	+0.83	+0.21
	NMAD''	0.023	0.110
	cat.	8%	42%
COSMOS-5	small- N wash (2/5 above our ceiling)		
ACS-13 (arc radius)	NMAD''	0.31	0.36
	no θ_E GT; excluded from aggregate		

$\theta_E < \text{arc-radius}$ negative bias. *Reproducibility caveat:* we cannot recover LEMON’s published aggregate (RMSE $0.14''$, NMAD $0.11''$, $R^2 = 0.53$) by scoring their released predictions against the literature θ_E ; we therefore report our transparent re-scoring separately from LEMON’s own reported numbers and flag the discrepancy [TODO: (R4: resolve with LEMON authors before submission)].

5.8 Transfer to Euclid and Roman

Transfer to a new survey proceeds by re-running the deterministic instrument operator (Section 3) to render the training population into the target domain, then retraining the same architecture and normalisation — no change to the network. For *Euclid* this is the GEN5 arm evaluated on real Q1 lenses [TODO: (E5)]. For Roman we render the GEN5 population into the WFI format: zero-shot transfer of the HST-trained model fails ($R^2 = 0.11$), whereas retraining on the Roman-rendered population succeeds ($R^2 = 0.93\text{--}0.94$). Since no real Roman lenses exist pre-launch, the Roman result is a simulation-to-simulation demonstration of instrument-agnostic retraining, with real validation deferred to post-launch (Section 6).

6 DISCUSSION

Self-consistency versus geometry. Our θ_E labels are $\text{SIS}(\sigma_v)$, but the network is not bottlenecked by them: the $\sigma_v \rightarrow \theta_E$ relation predicts the true b_{SIE} only to $\sim 17\%$, yet the CNN is four times tighter (Section 5.2). The self-consistency construction therefore supplies the training θ_E prior and a luminosity fallback for faint or ambiguous arcs, while the network’s precision on well-resolved systems is geometric. This delimits the claim honestly: self-consistency is decisive for the hard, faint-arc cases that break parametric simulators, not a universal substitute for resolving the arc.

Hard cases stay hard. Consistent with Prof. Chan’s observation, the network does not solve the intrinsically degenerate cases (faint arcs, out-of-support θ_E , complex environments); the confidence gate is designed to *flag* rather than fix them.

Calibration and scope. Raw uncertainties are mis-calibrated on real data and require recalibration (Section 5.5); the method is single-band and θ_E -only. We view θ_E as the most robust first target; ellipticity, shear, and source parameters are natural extensions.

Roman and the future. Real-Roman validation awaits launch; multi-band inputs, joint multi-parameter heads, and domain adaptation *on top of* realism-calibrated simulations are the clear next steps.

7 CONCLUSIONS

Physical realism in the training simulations — not architecture, and not domain adaptation — is the decisive factor in closing the sim-to-real gap for CNN Einstein-radius estimation. A real-ingredient, physically self-consistent generator, in which each deflector supplies both its light and (through its measured σ_v) its mass, lifts real-lens R^2 from strongly negative to +0.64–0.90, matching conventional modelling at a fraction of the amortized cost, with a domain-aware uncertainty that flags its own failures. The generator is instrument-agnostic and already validated across the HST and *Euclid* domains, with a clear path to Roman.

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DATA AVAILABILITY

The generator code, configuration files, kinematic library (g1b_kinematics_v1.csv), and per-lens predictions underlying this article are available at [TODO: repository/DOI]. The benchmark ground truth is from Bolton et al. (2008); COSMOS sources from Auger, Treu, Bolton, Gavazzi, Koopmans, Marshall, Moustakas & Burles (Man).

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APPENDIX A: REPRODUCIBILITY DETAILS

[TODO: seed lists, exact config filenames, gate thresholds.]

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